Transfer of methicillin-resistant Staphylococcus aureus from retail pork products onto food contact surfaces and the potential for consumer exposure.

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Methicillin-resistant Staphylococcus aureus (MRSA) is a pathogen that has developed resistance to beta-lactam antibiotics and has been isolated at low population numbers in retail meat products. The objectives of this study were to estimate the potential transfer of MRSA from contaminated retail pork products to food contact surfaces and to estimate the potential for human exposure to MRSA by contact with those contaminated surfaces. Pork loins, bacon, and fresh pork sausage were inoculated with a four-strain mixed MRSA culture over a range of populations from approximately 4 to 8 log, vacuum packaged, and stored for 2 weeks at 5°C to simulate normal packaging and distribution. Primary transfer was determined by placing inoculated products on knife blades, cutting boards, and a human skin model (pork skin) for 5 min. Secondary transfer was determined by placing an inoculated product on the contact surface, removing it, and then placing the secondary contact surface on the initial contact surface. A pork skin model was used to simulate transfer to human skin by placing it into contact with the contact surface. The percentages of transfer for primary transfer from the inoculated products to the cutting board ranged from 39 to 49%, while the percentages of transfer to the knife ranged from 17 to 42%. The percentages of transfer from the inoculated products to the pork skin ranged from 26 to 36%. The secondary transfer percentages ranged from 2.2 to 5.2% across all products and contact surfaces. Statistical analysis showed no significant differences in the amounts of transfer between transfer surfaces and across cell concentrations.